

Aditya Varshney

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Education

B.Tech – Computer Science (AIML & IoT)

GLA University, Mathura

2023 – 2027

Uttar Pradesh, India

CPI: 8.14 (*Till 6th Semester, Third Year Completed*)

Profile Summary

Fully experienced in developing full-stack software using JavaScript and with hands-on experience building production-level software using the full JavaScript stack ranging from back-end with Node.js to front-end with React as well as RESTful APIs used by actual customers. Shows a high level of ability to deploy front-end and back-end functionality, troubleshooting live problems, and explaining why certain choices were made architecturally. Has expert knowledge in Machine Learning, Deep Learning, and GenAI/NLP. Willing to work remotely and self-motivated.

Skills

Programming Languages: Python, Java, JavaScript (ES6+), SQL

DSA: Data Structures & Algorithms (Arrays, Linked List, Stack, Queue, Trees, Graphs, Recursion, Sorting, Searching, DP)

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Framer Motion

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Auth, WebSockets

Databases: MongoDB (Mongoose), MySQL

Data Science: EDA, Data Cleaning, Feature Engineering, Regression, Classification, Model Evaluation

Deep Learning: ANN, CNN, RNN, Regularization, Model Optimization

NLP: Tokenization, Text Preprocessing, Embeddings, Sentiment Analysis, RNN/LSTM basics

GenAI: LangChain, Groq API, Prompt Engineering, RAG, HuggingFace Transformers

AI / ML Libraries: scikit-learn, TensorFlow, OpenCV, Pandas, NumPy, Matplotlib, Seaborn

Tools & Platforms: Git, GitHub, Postman, VS Code, Jupyter Notebook, Google Colab

Projects

Real Time Video Analysis System

Python · OpenCV · Flask · CNN · JavaScript

- Built a real-time video processing pipeline using OpenCV and Python - handles live webcam streams and video file input with frame-by-frame deep learning inference using a CNN-based detection model.
- Designed a modular Flask REST API layer exposing inference endpoints, allowing the JavaScript frontend to display live annotated video feeds and real-time confidence scores on a responsive dashboard.
- Optimized the frame sampling and response cycle to reduce streaming latency - applied ROI extraction, confidence thresholding, and batch frame handling to maximize throughput without dropping stream quality.

ZENIX - AI Voice Assistant

React · FastAPI · Groq API · MongoDB · Python

- Developed a full-stack AI-based voice assistant with React and FastAPI. The system includes real-time voice processing, Groq-backed large language model outputs, and a dashboard UI with glassmorphism cards and Framer Motion animations.
- The backend extension was done in increments with the addition of components for emotion recognition, semantics deduplication, PII scrubbing and `ZenixOrchestrator` class implementation. The original service script was not modified and is fully compatible.
- Improved the FastAPI authentication layer with JWT, CSRF protection, custom token exception handlers and OWASP-compliant hashing of passwords - production-grade security from day one.
- Integrated MongoDB Atlas for persistent conversation storage and implemented caching and rate limiting to reduce redundant LLM calls.

InterviewVerse AI — Interview Prep Platform

React · Node.js · FastAPI — In Development

- Built the complete product frontend from scratch - landing page, admin/student login flows, dashboard with analytics, and a live interview session component with camera/mic access, real-time speech-to-text, and proctoring logic.
- Established a cohesive dark UI design system across all pages: glassmorphism panels, CSS grid overlays, radial orbs, and a deliberate three-color accent language (cyan, amber, emerald) for role-based visual identity.
- Resolved a production camera flickering bug in the interview session by tracing the root cause to React state re-renders on media stream re-initialization - applied targeted memoization and ref-based stream management.
- Integrated AI-powered question generation and answer evaluation APIs, delivering personalized feedback to students post-interview.

Certifications

- **Python + Machine Learning** | *Coding Blocks*
- **Advanced AIML Training** | *Pregrad Pvt. Ltd.*